EXPERIMENT 9: TITANIC DATASET ANALYSIS

AIM:

To analyze the Titanic dataset using Python by inspecting dataset information and summary statistics, handling missing values, visualizing passenger-class distribution, filtering passengers based on specified conditions, and splitting the data into training and testing sets.

ALGORITHM:

1. Import the required libraries: Pandas, Seaborn, Matplotlib, SimpleImputer, and train_test_split.

2. Load the Titanic CSV dataset into a Pandas DataFrame and display the first five rows.

3. Display the dataset shape, information, and summary statistics.

4. Fill missing values in the Age column using SimpleImputer with the mean strategy.

5. Replace missing values in the Cabin column with “Unknown”.

6. Fill missing values in the Embarked column using its mode.

7. Plot the passenger class distribution using a count plot.

8. Display female passengers who survived.

9. Display third-class passengers younger than 18 years.

10. Display first-class passengers older than 40 years.

11. Display surviving passengers in the specified category.

12. Display male passengers whose fare is greater than 100.

13. Display passengers who embarked from C and belonged to passenger class 2.

14. Display passengers with SibSp greater than 2.

15. Display non-surviving passengers who had no siblings/spouses and no parents/children aboard.

16. Display the five oldest surviving passengers.

17. Display passengers who paid zero fare.

18. Split the dataset into training and testing sets with test_size=0.2 and random_state=42.

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 1. Import Libraries and Load Titanic Dataset

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split

df = pd.read_csv(r"C:\\Users\\DELL\\Downloads\\archive (3)\\Titanic-Dataset.csv")

print("Dataset Loaded Successfully!")
print(df.head())
```

```
Dataset Loaded Successfully!
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 2. Understand the Data

print("Shape:", df.shape)

print("\nDataset Information")
df.info()

print("\nSummary Statistics")
print(df.describe())
```

```
Shape: (891, 12)

Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

Summary Statistics
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 3. Fill Missing Age Using SimpleImputer

imputer = SimpleImputer(strategy="mean")
df[["Age"]] = imputer.fit_transform(df[["Age"]])

print(df["Age"].isnull().sum())
```

```
0
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 4. Fill Missing Cabin with "Unknown"

df["Cabin"] = df["Cabin"].fillna("Unknown")

print(df["Cabin"].isnull().sum())
```

```
0
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 5. Fill Missing Embarked with Mode

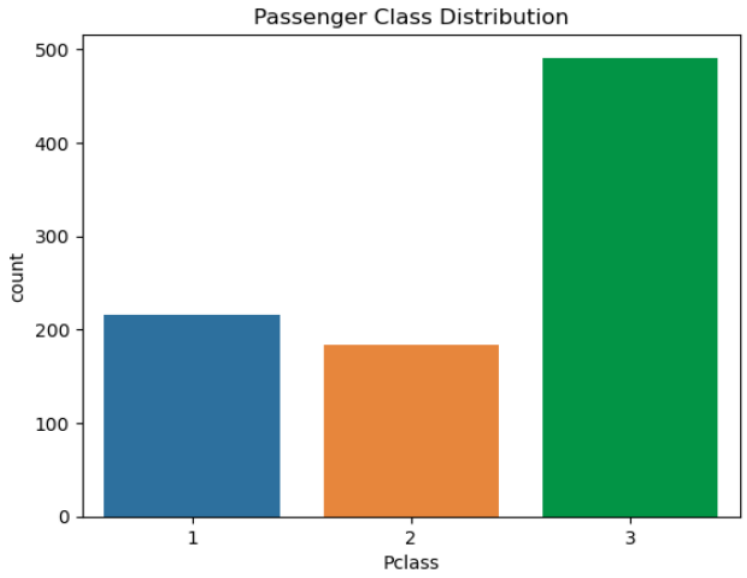
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

print(df["Embarked"].isnull().sum())
```

```
0
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 6. Passenger Class Count Plot

sns.countplot(x="Pclass", data=df)
plt.title("Passenger Class Distribution")
plt.show()
```




```
# EXERCISE-2 MUTHURAJ(241801176)
# 7. Female Passengers Who Survived

result = df[(df["Sex"] == "female") & (df["Survived"] == 1)]

print(result["Name"])
```

```
1      Cumings, Mrs. John Bradley (Florence Briggs Th...
2                                  Heikkinen, Miss. Laina
3           Futrelle, Mrs. Jacques Heath (Lily May Peel)
8      Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
9                    Nasser, Mrs. Nicholas (Adele Achem)
                             ...
874                Abelson, Mrs. Samuel (Hannah Wizosky)
875                     Najib, Miss. Adele Kiamie "Jane"
879        Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)
880         Shelley, Mrs. William (Imanita Parrish Hall)
887                           Graham, Miss. Margaret Edith
Name: Name, Length: 233, dtype: object
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 8. 3rd Class Passengers Under Age 18

result = df[(df["Pclass"] == 3) & (df["Age"] < 18)]

print(result["Name"])
```

```
7                Palsson, Master. Gosta Leonard
10             Sandstrom, Miss. Marguerite Rut
14         Vestrom, Miss. Hulda Amanda Adolfina
16                         Rice, Master. Eugene
22                 McGowan, Miss. Anna "Annie"
                          ...
844                        Culumovic, Mr. Jeso
850    Andersson, Master. Sigvard Harald Elias
852                    Boulos, Miss. Nourelain
869             Johnson, Master. Harold Theodor
875           Najib, Miss. Adele Kiamie "Jane"
Name: Name, Length: 78, dtype: object
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 9. Class 1 Passengers Older Than 40

result = df[(df["Pclass"] == 1) & (df["Age"] > 40)]

print(result["Name"])
```

```
6                                 McCarthy, Mr. Timothy J
11                               Bonnell, Miss. Elizabeth
35                          Holverson, Mr. Alexander Oskar
52              Harper, Mrs. Henry Sleeper (Myna Haxtun)
54                          Ostby, Mr. Engelhart Cornelius
                                ...
856           Wick, Mrs. George Dennick (Mary Hitchcock)
857                                Daly, Mr. Peter Denis
862    Swift, Mrs. Frederick Joel (Margaret Welles Ba...
871     Beckwith, Mrs. Richard Leonard (Sallie Monypeny)
879        Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)
Name: Name, Length: 76, dtype: object
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 10. Survived Passengers in Above Category

result = df[
    (df["Pclass"] == 1) &
    (df["Age"] > 40) &
    (df["Survived"] == 1)
]

print(result["Name"])
```

```
11                               Bonnell, Miss. Elizabeth
52              Harper, Mrs. Henry Sleeper (Myna Haxtun)
187        Romaine, Mr. Charles Hallace ("Mr C Rolmane")
194            Brown, Mrs. James Joseph (Margaret Tobin)
195                                  Lurette, Miss. Elise
268        Graham, Mrs. William Thompson (Edith Junkins)
275                     Andrews, Miss. Kornelia Theodosia
299      Baxter, Mrs. James (Helene DeLaudeniere Chaput)
337                       Burns, Miss. Elizabeth Margaret
366     Warren, Mrs. Frank Manley (Anna Sophia Atkinson)
380                                  Bidois, Miss. Rosalie
449                         Peuchen, Major. Arthur Godfrey
453                               Goldenberg, Mr. Samuel L
460                                    Anderson, Mr. Harry
496                         Eustis, Miss. Elizabeth Mussey
513       Rothschild, Mrs. Martin (Elizabeth L. Barrett)
523      Hippach, Mrs. Louis Albert (Ida Sophia Fischer)
556    Duff Gordon, Lady. (Lucille Christiana Sutherl...
571        Appleton, Mrs. Edward Dale (Charlotte Lamson)
587                      Frolicher-Stehli, Mr. Maxmillian
591      Stephenson, Mrs. Walter Bertram (Martha Eustis)
599         Duff Gordon, Sir. Cosmo Edmund ("Mr Morgan")
621                           Kimball, Mr. Edwin Nelson Jr
630                  Barkworth, Mr. Algernon Henry Wilson
645                                Harper, Mr. Henry Sleeper
647                  Simonius-Blumer, Col. Oberst Alfons
660                        Frauenthal, Dr. Henry William
707                     Calderhead, Mr. Edward Pennington
712                                Taylor, Mr. Elmer Zebley
765                  Hogeboom, Mrs. John C (Anna Andrews)
779    Robert, Mrs. Edward Scott (Elisabeth Walton Mc...
796                            Leader, Dr. Alice (Farnham)
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 11. Male Passengers Who Paid Fare Greater Than 100

result = df[(df["Sex"] == "male") & (df["Fare"] > 100)]

print(result[["Name", "Fare"]])
```

```
                                     Name      Fare
27                Fortune, Mr. Charles Alexander  263.0000
118                     Baxter, Mr. Quigg Edmond  247.5208
305               Allison, Master. Hudson Trevor  151.5500
332                    Graham, Mr. George Edward  153.4625
373                          Ringhini, Mr. Sante  135.6333
377                    Widener, Mr. Harry Elkins  211.5000
390                  Carter, Mr. William Ernest  120.0000
438                             Fortune, Mr. Mark  263.0000
505  Penasco y Castellana, Mr. Victor de Satode  108.9000
527                             Farthing, Mr. John  221.7792
544                    Douglas, Mr. Walter Donald  106.4250
550                  Thayer, Mr. John Borland Jr  110.8833
557                             Robbins, Mr. Victor  227.5250
659                    Newell, Mr. Arthur Webster  113.2750
660                 Frauenthal, Dr. Henry William  133.6500
679           Cardeza, Mr. Thomas Drake Martinez  512.3292
698                      Thayer, Mr. John Borland  110.8833
737                        Lesurer, Mr. Gustave J  512.3292
802          Carter, Master. William Thornton II  120.0000
```

```
# EXERCISE-2 MUTHURAJ(241801176)
# 12. Passengers Embarked from C and Class 2

result = df[(df["Embarked"] == "C") & (df["Pclass"] == 2)]

print(result["Name"])
```

```
9                      Nasser, Mrs. Nicholas (Adele Achem)
43               Laroche, Miss. Simonne Marie Anne Andree
122                                 Nasser, Mr. Nicholas
135                                   Richard, Mr. Emile
181                                     Pernot, Mr. Rene
292                                 Levy, Mr. Rene Jacques
308                                    Abelson, Mr. Samuel
361                               del Carlo, Mr. Sebastiano
389                                   Lehmann, Miss. Bertha
473           Jerwan, Mrs. Amin S (Marie Marthe Thuillard)
547                               Padro y Manent, Mr. Julian
608    Laroche, Mrs. Joseph (Juliette Marie Louise La...
685                 Laroche, Mr. Joseph Philippe Lemercier
817                                     Mallet, Mr. Albert
827                                  Mallet, Master. Andre
866                            Duran y More, Miss. Asuncion
874                  Abelson, Mrs. Samuel (Hannah Wizosky)
Name: Name, dtype: object
```

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 13. Passengers with SibSp > 2

result = df[df["SibSp"] > 2]

print(result[["Name", "SibSp"]])
```

```
                                                  Name  SibSp
7                            Palsson, Master. Gosta Leonard      3
16                                    Rice, Master. Eugene      4
24                             Palsson, Miss. Torborg Danira      3
27                           Fortune, Mr. Charles Alexander      3
50                               Panula, Master. Juha Niilo      4
59                     Goodwin, Master. William Frederick      5
63                                  Skoog, Master. Harald      3
68                        Andersson, Miss. Erna Alexandra      4
71                             Goodwin, Miss. Lillian Amy      5
85   Backstrom, Mrs. Karl Alfred (Maria Mathilda Gu...      3
88                             Fortune, Miss. Mabel Helen      3
119                    Andersson, Miss. Ellis Anna Maria      4
159                            Sage, Master. Thomas Henry      8
164                        Panula, Master. Eino Viljami      4
171                                   Rice, Master. Arthur      4
176                         Lefebre, Master. Henry Forbes      3
180                          Sage, Miss. Constance Gladys      8
182                Asplund, Master. Clarence Gustaf Hugo      4
201                                   Sage, Mr. Frederick      8
229                              Lefebre, Miss. Mathilde      3
233                         Asplund, Miss. Lillian Gertrud      4
261                    Asplund, Master. Edvin Rojj Felix      4
266                             Panula, Mr. Ernesti Arvid      4
278                                    Rice, Master. Eric      4
324                              Sage, Mr. George John Jr      8
341                           Fortune, Miss. Alice Elizabeth      3
374                             Palsson, Miss. Stina Viola      3
386                          Goodwin, Master. Sidney Leonard      5
409                                   Lefebre, Miss. Ida      3
480                          Goodwin, Master. Harold Victor      5
485                              Lefebre, Miss. Jeannie      3
541                 Andersson, Miss. Ingeborg Constanzia      4
542                    Andersson, Miss. Sigrid Elisabeth      4
634                                   Skoog, Miss. Mabel      3
```

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 14. Non-Survivors with No Family Aboard

result = df[
    (df["Survived"] == 0) &
    (df["SibSp"] == 0) &
    (df["Parch"] == 0)
]

print(result["Name"])
```

```
4                       Allen, Mr. William Henry
5                              Moran, Mr. James
6                        McCarthy, Mr. Timothy J
12                Saundercock, Mr. William Henry
14         Vestrom, Miss. Hulda Amanda Adolfina
                          ...
882                 Dahlberg, Miss. Gerda Ulrika
883                Banfield, Mr. Frederick James
884                       Sutehall, Mr. Henry Jr
886                        Montvila, Rev. Juozas
890                          Dooley, Mr. Patrick
Name: Name, Length: 374, dtype: object
```

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 15. Top 5 Oldest Passengers Who Survived

result = df[df["Survived"] == 1].nlargest(5, "Age")

print(result[["Name", "Age"]])
```

```
                                      Name   Age
630       Barkworth, Mr. Algernon Henry Wilson  80.0
275          Andrews, Miss. Kornelia Theodosia  63.0
483                        Turkula, Mrs. (Hedwig)  63.0
570                            Harris, Mr. George  62.0
829  Stone, Mrs. George Nelson (Martha Evelyn)  62.0
```

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 16. Passengers Who Paid Zero Fare

result = df[df["Fare"] == 0]

print(result[["Name", "Fare"]])
```

```
                                 Name  Fare
179               Leonard, Mr. Lionel   0.0
263             Harrison, Mr. William   0.0
271       Tornquist, Mr. William Henry   0.0
277        Parkes, Mr. Francis "Frank"   0.0
302    Johnson, Mr. William Cahoone Jr   0.0
413     Cunningham, Mr. Alfred Fleming   0.0
466             Campbell, Mr. William   0.0
481  Frost, Mr. Anthony Wood "Archie"   0.0
597               Johnson, Mr. Alfred   0.0
633     Parr, Mr. William Henry Marsh   0.0
674        Watson, Mr. Ennis Hastings   0.0
732              Knight, Mr. Robert J   0.0
806            Andrews, Mr. Thomas Jr   0.0
815                  Fry, Mr. Richard   0.0
822   Reuchlin, Jonkheer. John George   0.0
```

```python
# EXERCISE-2 MUTHURAJ(241801176)
# 17. Split Dataset for Training and Testing

train_data, test_data = train_test_split(
    df,
    test_size=0.2,
    random_state=42
)

print("Training Data:", train_data.shape)
print("Testing Data:", test_data.shape)
```

```
Training Data: (712, 12)
Testing Data: (179, 12)
```

RESULT:

The Titanic dataset was successfully loaded and analyzed. Dataset information and summary statistics were displayed, missing values in Age, Cabin, and Embarked were handled, and passenger-class distribution was visualized. Passenger records were filtered using the specified conditions. Finally, the dataset was split into 712 training records and 179 testing records.